

Literature Review



Literature Review

Part B



Literature Review

- Is an integral part of the research process
- makes a valuable contribution to almost every operational step.
- Has value even before the first step (that is, when you are merely thinking about a research question that you may want to find answers to through your research journey).
- Helps to establish the theoretical roots of your study.
- Serves to enhance and consolidate your own knowledge base and helps to integrate your findings with the existing body of knowledge.



Function of LR

- Bring clarity and focus to your research problem.
- Improve your research methodology.
- Broaden your knowledge base in your research area.
- Contextualize your findings.



CREATING OUTLINE AND REVIEWING LITERATURE

- After the previous steps have been completed it is important for the researcher to analyse approaches that help with the foundation of the topic.
- Usually observed as an index with titles and subtitles.
- It is important to know that looking over at the existent literature helps out with figuring out the best way, and more importantly, the manner in which we want to deal with our research and the topic.



THEORY

- Theory: actual knowledge that helps understand events, situations and context. It describes, explains, predicts and finally orients the research process.
 - Criteria to evaluate:
 - Able to explain, describe and predict
 - Logical consistency
 - Perspective: that it explains the topic (almost) totally
 - Innovation: capability to generate new discoveries
 - Simplicity: not required but desired in order to easily understand it
-

What is a Literature Search?

- Identification of what has already been written on a topic of interest
- Investigation of literature on research methodology, where appropriate.
- A basis for judging the feasibility of a proposed project (Based on what is known, is it reasonable to assume whether the objectives of the project can be achieved?)



LITERATURE SURVEY



Documentation of a comprehensive review of the published and unpublished work from secondary sources of data in the area of specific interest to the researcher.



For example:

Journals, conference proceedings, books, doctoral dissertations, government publication, financial and marketing reports, etc..

- Selection of useful and **accurate keywords** relating to the topic of interest
- Identification of **high-quality resources** containing information on the topic of interest (Is the resource reliable?)
- Retrieval of information
- Effective management of searches (keeping track of what you find)

Main Elements of a Literature Search

A good literature survey

- Ensure that **important variables** are not left out of the study.
- Helps the development of the theoretical framework and hypotheses for testing.
- Ensures that the **problem statement is precise and clear.**
- Enhances **testability and replicability** of the findings
- Reduces the risk of “reinventing the wheel”
- Confirms that the problem is perceived as relevant and significant

LITERATURE SURVEY



Conducting Literature Review (Ranjit Kumar, 2018)

There are 4 steps involved in conducting a literature review

- Searching for the existing literature in your area of study.
- Reviewing the selected literature.
- Developing a theoretical framework.
- Developing a conceptual framework.

CONDUCTING LITERATURE REVIEW (LR)

Data Sources

Searching for literature

Evaluating the literature

Documenting the LR

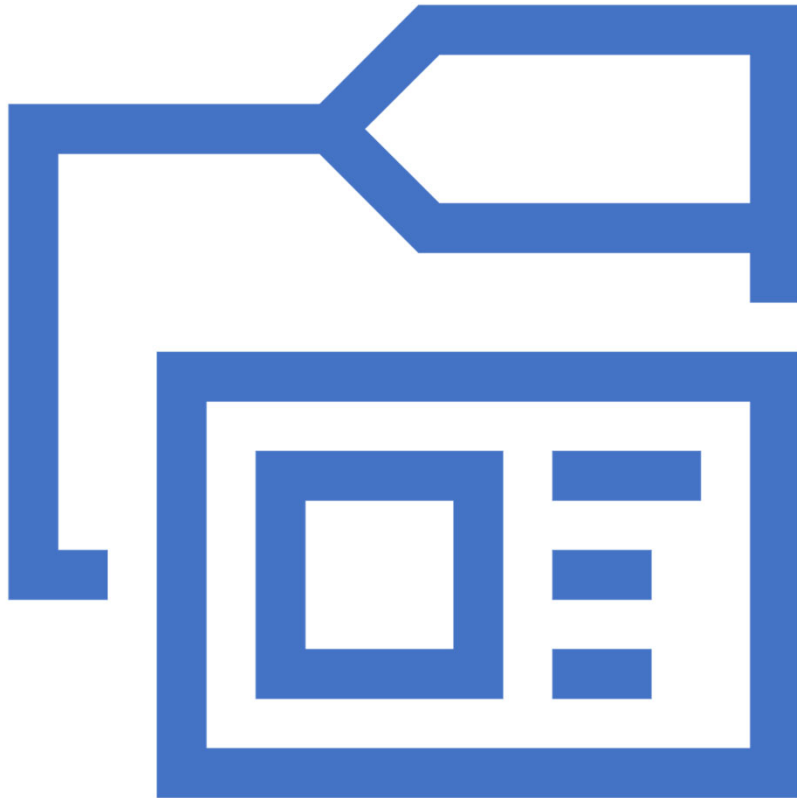
DATA SOURCES

- Journal articles
- Books
- Conference proceedings
- Government and corporate reports
- Newspapers
- Theses and dissertations
- Internet (electronic journals)
- CD-ROM (old time)
- Magazines



JOURNAL ARTICLES

- These are good especially for **up-to-date information**. Bear in mind, though, that it can take up to two years to publish articles. There are frequently used in literature reviews because they offer a **relative concise, up – to date format for research** and because all reputable journals are refereed (such that, editor publish only the most relevant and reliable research)



BOOKS

- Books tend to be **less up-to date** as it takes longer for a book to be published than for a journal article. **Text books are unlikely to be useful for LR as they are intended for teaching, not for research.** But they do offer a good starting point from which to find more detailed sources.



- These can be useful in providing the latest research or research that has not been published. They are also helpful in providing information on which people are currently involved in which research areas and so can be helpful in tracking down other work by the same researchers.

CONFERENCE PROCEEDINGS

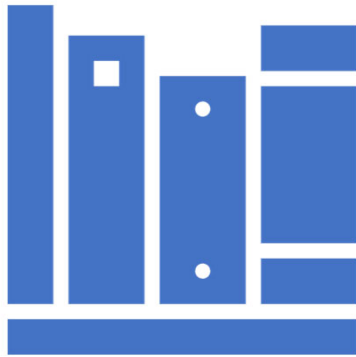
GOVERNMENT/CORPORATE REPORTS



- Many government departments and corporations commission or carry out research. their published findings can provide a useful sources of information depending on your filed of study.

NEWSPAPER

- Since newspapers are generally intended for a general (not specialized) audience, the information they provide will be of **very limited use for your literature review**. Often newspapers are more helpful **as providers of information about recent trends, discoveries or changes**.
- For example, announcing changes in government policy, but you should then search for more detailed information in other sources.



- These can be useful sources of information. However there are disadvantages:
 - They can be difficult to obtain since they are not published, but are generally only available from the library shelf or through interlibrary loan.
 - The student who carried out the research may not be an experienced researcher and therefore you might have to treat their findings with more caution than published research

THESES AND DISSERTATIONS



INTERNET

- The fastest-growing source of information is on the internet. It is impossible to characterize the information available but here are some hints about using electronic sources:
 - Bear in mind that anyone can post information on the internet so the quality may not be reliable.
 - The information you find may be intended for a general audience and so not be suitable for inclusion in your literature review (information for a general audience is usually less detailed)
 - More and more refereed electronic journal (e-journals) appearing on the internet,
 - If they are refereed it means that there is an editorial board that evaluates the work before publishing it in their e-journal, so the quality should be more reliable (depending on the reputation of the journal).

CD-ROMS

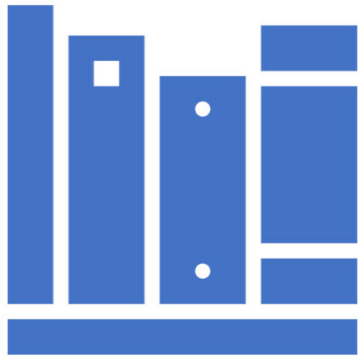


- At the moment, few CD-ROMS provide the kind of specialized, detailed information about academic research that you need for your own research since most are intended for a general audience.
- However, more and more bibliographies are being put onto CD-ROM for use in academic libraries, so they can be a very valuable tool in searching for the information you need.



MAGAZINES

- Magazines intended for a general audience (eg time) are unlikely to be useful in providing the sort of information you need.
- Specialized magazines may be more useful (for example business magazines for management students) but usually magazines **are not useful for your research except as a starting point by providing news or general information** about new discoveries, policies, etc.. That you can further research in more specialized sources



- Example:
- Your library search assignment

SEARCHING FOR LITERATURE

**Your
topic**

UNDERSTAND

Identify keywords and concepts

WHERE TO LOOK

Library, Databases, WWW, DDS

EVALUATE

Quality, Evidence

Search strategies



Literature Searching & Review (how it starts)





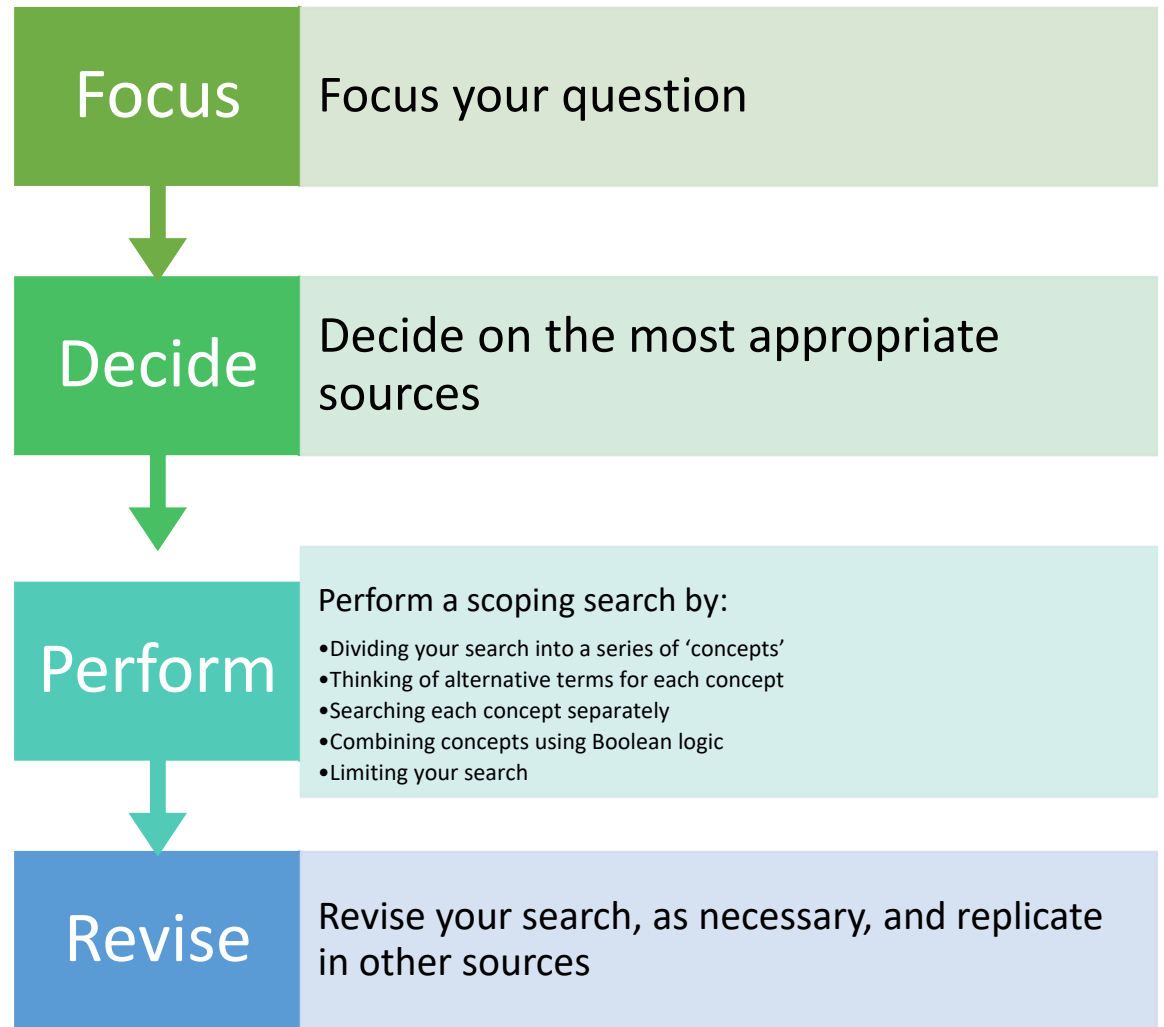
Literature Searching & Review (How it might end up)



Literature Search techniques

- These will vary depending on where the source you are searching, and you should always investigate what advanced search techniques are offered by the individual sources as they will not be consistent.
- Below are some of the most regular search techniques available.
 - **Keywords**
 - **Boolean searching (AND, OR, NOT)**
 - **Combining words**
 - **Phrase searching**
 - **Truncation** (e.g Nurs* = nurse, nurses, nursing, nursery)
or **wildcard searching** (e,g wom*n with retrieve the terms woman and women).
 - **Focusing a search** (use field tags or limit your search)

Stages in the literature search process



Keywords in Literature Search

- Think around the topic as much as possible and record all possible terms that could be used. This is essential as two sources may use two different terms to define the same topic.
- Consider the following issues:
 - synonyms - e.g. cancer and neoplasm
 - acronyms
 - alternative spellings - try UK and US spellings (e.g. paediatric and pediatric)
 - alternative and related terms
 - international terms
 - former terms (i.e. non pc versions)
 - changes of place name
 - broader and narrower terms - you must be sure your results will be pitched at the right level and broader terms can retrieve very general results while narrow terms may be too specific

A structure for formulated questions

Health services research uses **PATIENT-INTERVENTION-COMPARISON-OUTCOME** (PICO) structure

Within social sciences research the following may be more appropriate:

SETTING

POPULATION

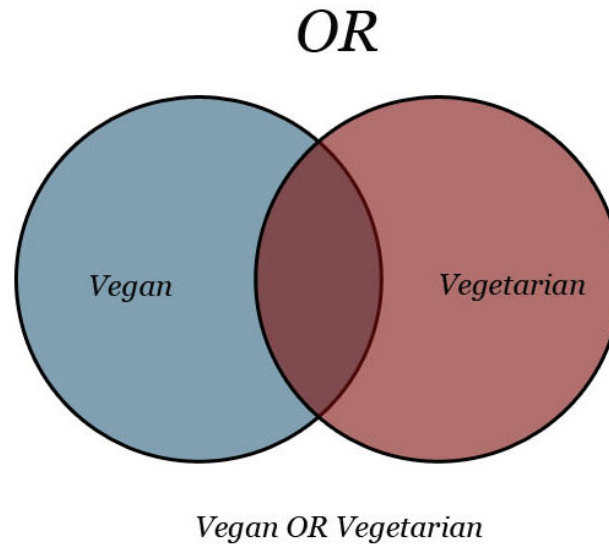
INTERVENTION

COMPARISON

EVALUATION

Boolean - OR

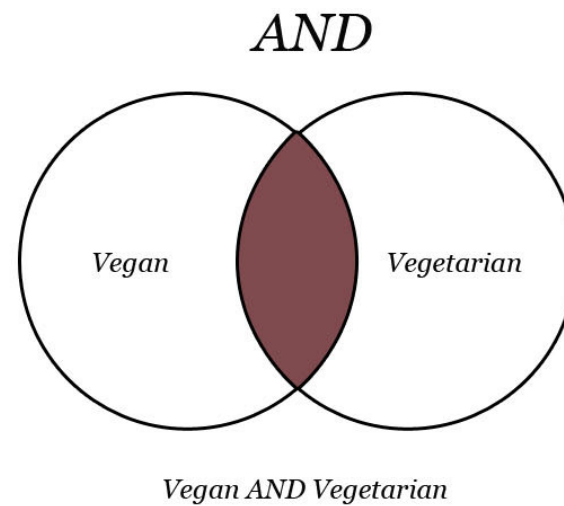
Use to combine like terms or terms within the same concept



Using OR, the student can find resources that include information about vegan diets, vegetarian diets, and resources that discuss both.

Boolean - AND

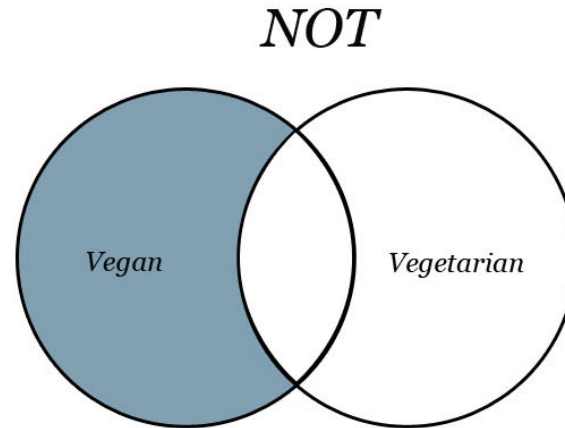
Use to combine together different concepts



Using AND, the student can find resources that include information about both vegan and vegetarian diets.

Boolean - NOT

Use to exclude terms from your search



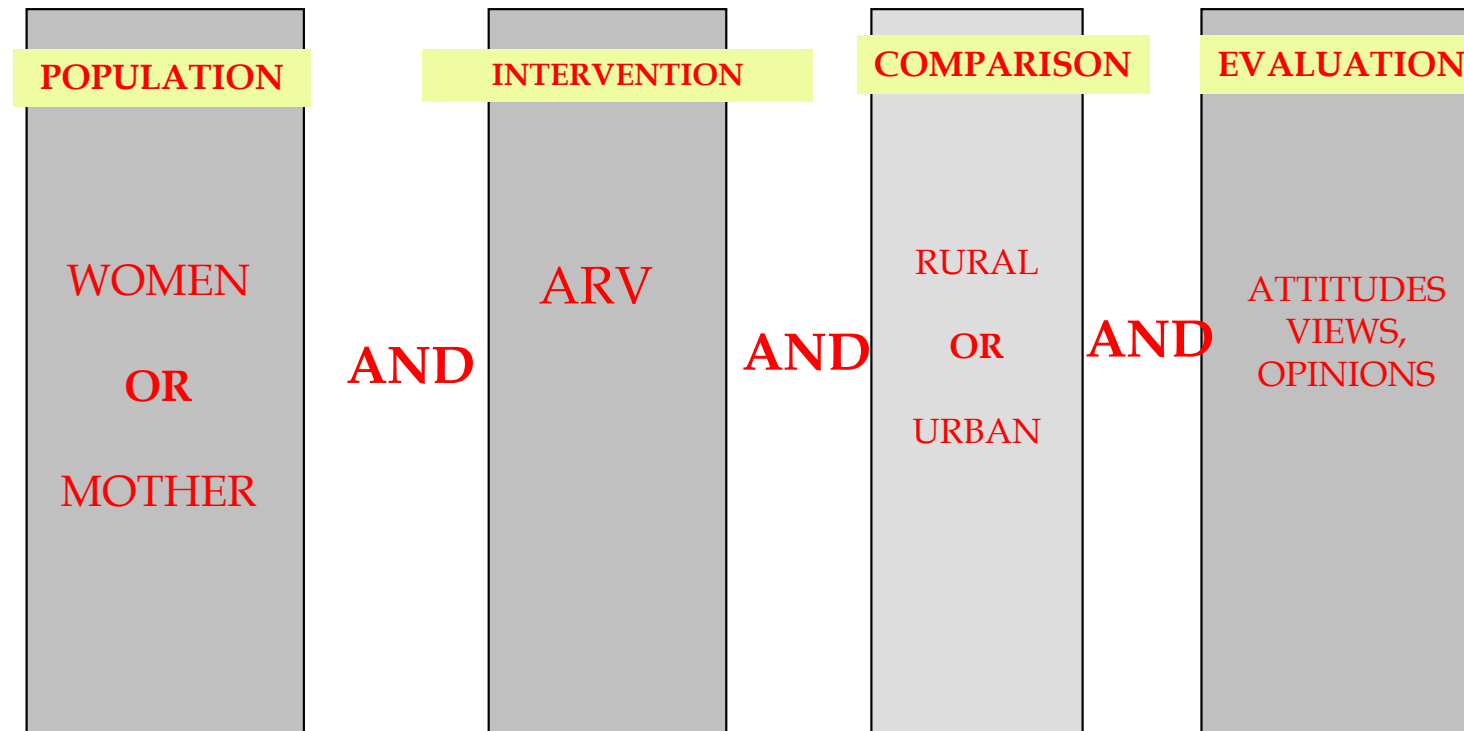
Vegan NOT Vegetarian

*Using NOT, the student can find
resources that only discuss vegan diets.*

An example of SPICE [Actual PhD example]

- **SETTING** – Malaysia
- **POPULATION** – Women or mothers
- **INTERVENTION** – anti-retroviral therapy
- **COMPARISON** – [rural and urban]
- **EVALUATION** – historical, political, attitudes, views, opinions

Using Boolean in your strategy





Organisation and systematic search

- A systematic literature search can be more easily performed if **the incoming material is kept well organized**
 - What keywords have I used so far?
 - Can I quickly gather all the papers I found on a certain topic?
 - When was the last found paper with a given keyword?
 - What are the catalogues I have used so far?
 - How many papers did I find on a certain topic?
 - Can I quickly retrieve a paper with a certain title, author?
 - Considering the result so far, what and how should I search more?

	Relevant	Not Relevant
Retrieved	Use to generate additional search terms	Eliminate terms with poor yield
Not Retrieved	Use reference lists from key articles to identify these	No further action required

Evaluating
your search
strategy

To optimize search on literatures



Must know the basic
understanding of



Databases

How the literature
is organized or
indexed

How different
databases differ

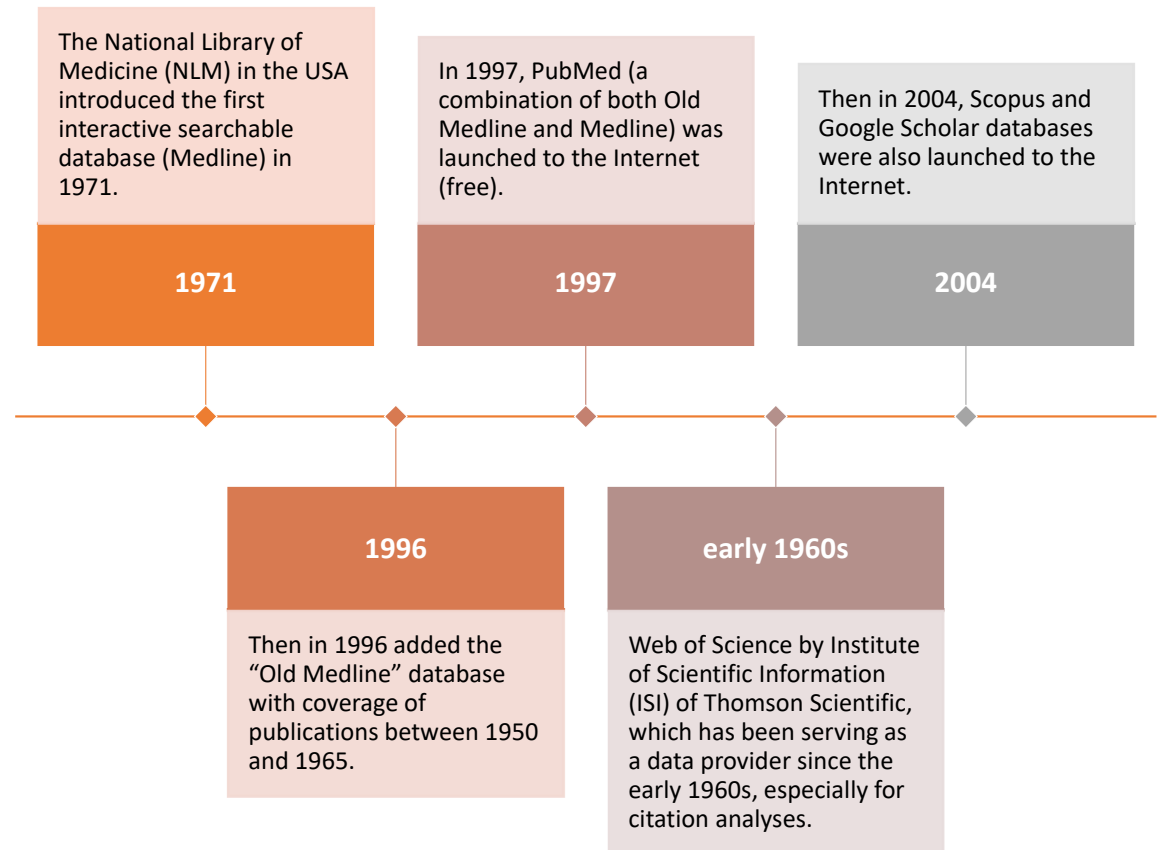


Search Engines

How search
engines different
from databases

How different
search engines
work

History of literature databases



Online Databases & Search engines

DATABASES

General

- Index Medicus (Medline)
- Web of Science
- EMBASE
- Cochrane

Specific

- PsychInfo
- CANCERLIT
- CINAHL
- AMED
- CIRRIE

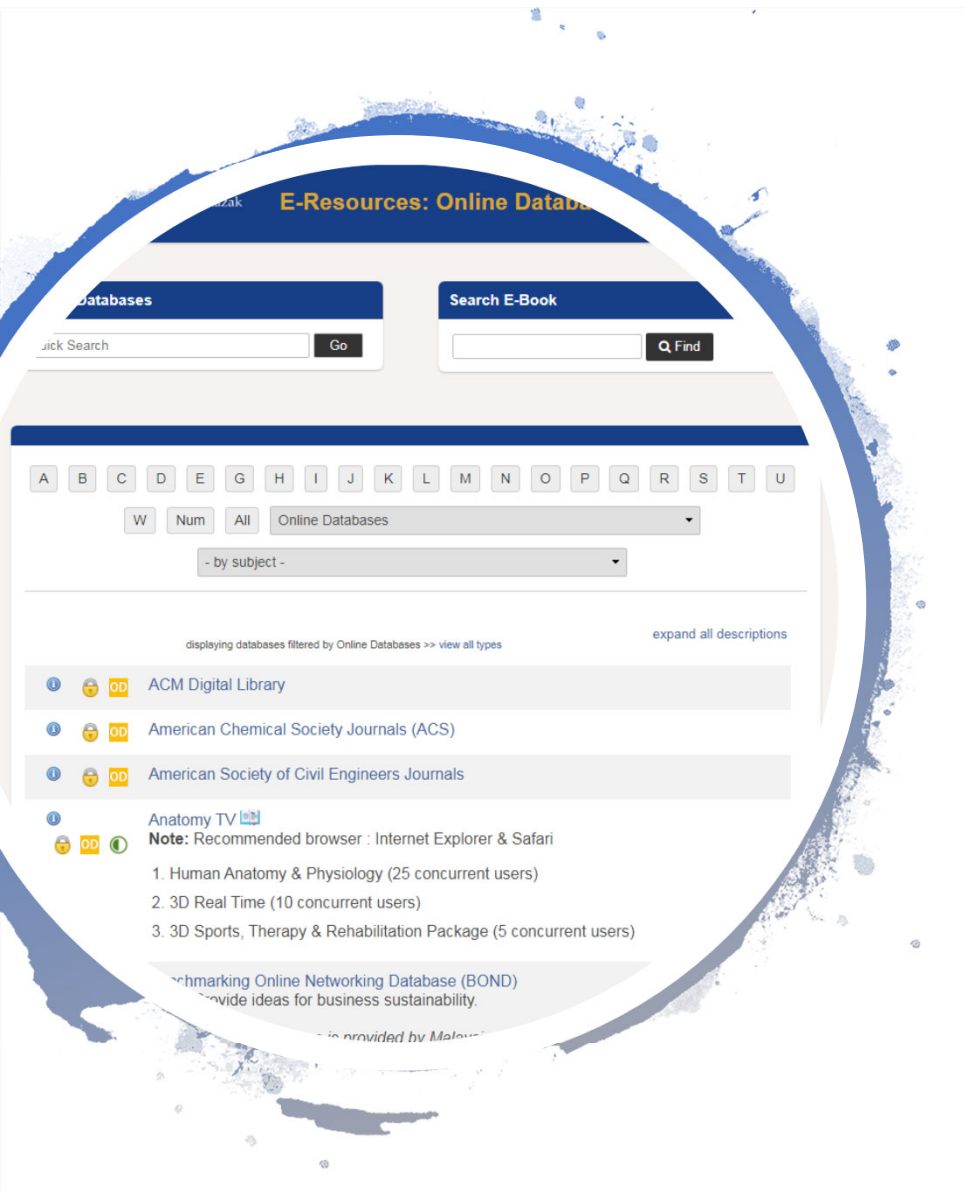
SEARCH ENGINES

General

- Google
- Bing
- Yahoo! Search
- Walfram alpha

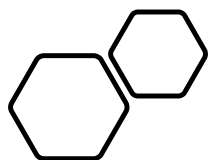
Specific

- PubMed
- Google Scholar
- Scirus
- Ovid, Proquest etc



UiTM subscribed-databases

- If you're not within UiTMNet, please click <http://ezaccess.library.uitm.edu.my>
- The content of these electronic resources are copyrighted and licensed material. Access is limited to ONLY authorized users.
- Use your **Library Username or Matric/Staff No & your New IC No.** for password to gain access.



Elsevier

- European-based products and services include electronic and print versions of journals, monographs, textbooks and reference works and cover the health, life, physical and social sciences.
- Organized in two divisions: Science & Technology and Health Sciences.



Web-based search engines

- Many proprietary medical resources and databases are publicly searchable by means of any Internet search engine. Access to the content, however, usually requires a subscription to the publication.
- **Google** and **Bing** are general search engines.
- **PubMed** and **Google Scholar** primarily search engines that find mostly journal articles, the former from the life sciences, the latter from all areas of research.

Main search engines for medical/health or life sciences literatures



PubMed



Google Scholar



Web of Science



Scopus

Characteristic	Pub Med	Scopus	Web of Science	Google Scholar
Date of official inauguration	06/1997 ^a	11/2004	2004 ^b	11/2004
Content				
No. of journals	6000 (827 open access)	12,850 (500 open access)	8700	No data provided (theoretically all electronic resources)
Languages	English (plus 56 other languages)	English (plus more than 30 other languages)	English (plus 45 other languages)	English (plus any language)
Focus (field)	Core clinical journals, dental journals, nursing journals, biomedicine, medicine, history of medicine, bioethics, space, life sciences	Physical sciences, health sciences, life sciences, social sciences	Science, technology, social sciences, arts and humanities	Biology, life sciences and environmental sciences, business, administration, finance and economics, chemistry and materials science, engineering, pharmacology, veterinary science, social sciences, arts and humanities
Period covered	1950–present	1966–present	1900–present	Theoretically all available electronically
Databases covered	Medline (1966–present), old Medline (1950–1965), PubMed Central, linked to other, more specialized, NLM databases	100% Medline, Embase, Compendex, World textile index, Fluidex, Geobase, Biobase	Science citation index expanded, social sciences citation index, arts and humanities citation index, index chemistry, current chemical reactions	PubMed, OCLC First Search
No. of keywords allowed	No limit	30	15	Theoretically no limit

PubMed

<https://www.ncbi.nlm.nih.gov/pmc/>

<https://pubmed.ncbi.nlm.nih.gov/>



A very handy, quick and easy to use database.



PubMed now offers 1 million freely available articles of which a significant number come from digitized back issues.



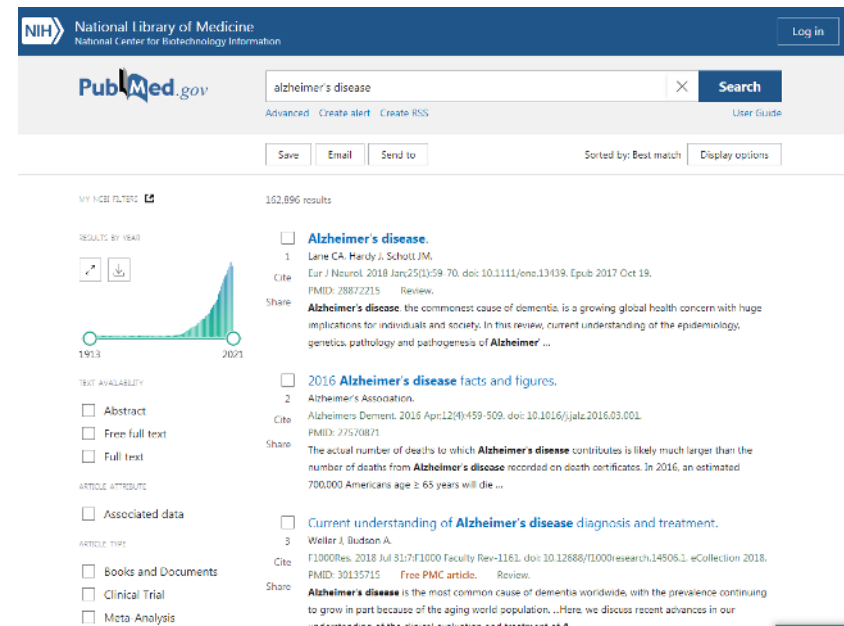
PubMed searches define databases that have extensive indexing and quality control.



In addition to searching the text for particular words, it uses a controlled-vocabulary thesaurus of medical subject headings, known as MeSH. This feature permits searching with various degrees of specificity.

What PubMed does not cover

- About 60 – 70% of all journals.
- Many pharmacology and therapeutics journals
- Many European journals
- Most journals from developing countries



Refining the search Search Field Tags in PubMed

Affiliation [ad]

Author Name [au]

Date of publication [dp]

Journal title [ta]

Language [la]

MeSH Heading [mh]

MeSH Major Topic [majr]

Publication type [pt]

Subheadings [sh]

Title Words [ti]

Title/Abstract Words [tiab]

Google Scholar vs PubMed

- Google Scholar is more difficult to focus than PubMed — but it may find papers, theses, books, preprints, abstracts, and technical reports that are not in National Library of Medicine databases.
- However, Google Scholar does not identify the sources that it has — or has not — indexed. Thus, there is no way to know what may be missing.

Google Scholar



Widely cited and important papers in a given field will often appear at the top of the results, with newer papers and others with fewer citations appearing near the end.



Searches can be limited.



The results take into account the full text of the article, the author, the publication in which it appeared, and how often it has been cited in other scholarly publications.



Google Scholar also includes a “cited by” feature that links an article to the others that have cited it.

Google Scholar

 Articles About 15,000 results (0.07 sec)

Any time

Since 2020

Since 2019

Since 2016

Custom range...

Sort by relevance

Sort by date

☒ include patents

☒ include citations

 Create alert

Blood biomarkers: Democratizing alzheimer's diagnostics
H Zetterberg, K Blennow - *Neuron*, 2020 - Elsevier
... Plasma P-tau181 concentration differentiated AD from non-AD neurodegenerative **diseases** and cognitively normal controls, was ... RO Roberts, MM Machulda, et al. Plasma phospho-tau181 increases with **Alzheimer's disease** clinical severity ... *Alzheimers Dement.*, 14 (2018), pp ...
  Cited by 1 All 6 versions

[HTML] Screening and Identification of Potential Peripheral Blood Biomarkers for Alzheimer's Disease Based on Bioinformatics Analysis
X Wang, L Wang - ... *Science Monitor: International Medical Journal of* ..., 2020 - ncbi.nlm.nih.gov
Background **Alzheimer's disease** (AD) is the leading cause of dementia worldwide; however, the molecular mechanisms underlying its pathogenesis remain unclear. The present study aimed to discover some potential peripheral **blood biomarkers** for early detection of patients ...
  All 5 versions

[HTML] Identification of potential blood biomarkers for early diagnosis of Alzheimer's disease through RNA sequencing analysis
D Shigemizu, T Mori, S Akiyama... - *Alzheimer's* ..., 2020 - alzres.biomedcentral.com
With demographic shifts toward older populations, the number of people with dementia is steadily increasing. **Alzheimer's disease** (AD) is the most common cause of dementia, and no curative treatment is available. The current best strategy is to delay **disease** progression ...
  Cited by 2 All 9 versions 

Blood biomarkers of traumatic brain injury and cognitive impairment in older veterans
CB Peltz, K Kenney, J Gill, R Diaz-Arrastia... - *Neurology*, 2020 - AAN Enterprises
Skip to main content. Advertisement. Neurology.org; Journals: Neurology; Clinical Practice; Genetics; Neuroimmunology & Neuroinflammation. Specialty Sites: COVID-19; Practice Current; Practice Buzz; Without Borders; Equity, Diversity ...
  Cited by 2 All 7 versions

Scopus

- The Scopus database was developed by Elsevier (Europe), combining the characteristics of both PubMed and Web of Science.
- Scopus includes a more expanded spectrum of journals than PubMed and Web of Science.
- Its citation analysis is faster and includes more articles than the citation analysis of Web of Science.
- On the other hand, the citation analysis that Web of Science presents provides better graphics and is more detailed than the citation analysis of Scopus.

<https://www.scopus.com/home.uri>

(Need to register)



Scopus

SearchSourcesListsSciVal   

Create account

Sign in

3,975 document results

TITLE-ABS-KEY (alzheimer's AND disease AND blood AND biomarkers)

 Edit  Save  Set alert

Search within results... 

Refine results

Limit to

Exclude

Access type  

☐ Open Access (1,530) >

☐ Other (2,445) >

Year 

☐ 2021 (2) >

☐ 2020 (432) >

☐ 2019 (568) >

☐ 2018 (441) >

☐ 2017 (437) >

View more

DocumentsSecondary documentsPatents

View Mendeley Data (974)

 Analyze search results

Show all abstracts

 Sort on:

Date (newest) 

☐ All 

Export

Download

View citation overview

View cited by

Add to List

...







	Document title	Authors	Year	Source	Cited by
☐ 1	Sleep-disordered breathing and the risk of Alzheimer's disease	Liguori, C., Maestri, M., Spanetta, M., (...), Mercuri, N.B., Guarnieri, B.	2021	Sleep Medicine Reviews 55,101375	0
View abstract  View at Publisher Related documents					
☐ 2	Genome-wide integrative analysis reveals common molecular signatures in blood and brain of alzheimer's disease	Rahman, M.R., Islam, T., Shahjaman, M., (...), Gov, E., Moni, M.A.	2021	Biointerface Research in Applied Chemistry 11(2), pp. 8686-8701	0
View abstract  View at Publisher Related documents					

Web of Science

Use of the Web of Science is licensed to institutions such as universities and the research departments of large corporations, generally through access to the Web of Knowledge.

The screenshot displays the Web of Science interface. At the top, a navigation bar includes links to 'Web of Science', 'InCites', 'Journal Citation Reports', 'Essential Science Indicators', 'EndNote', 'Publons', 'Kopernio', and 'Master Journal List'. On the right, there are links for 'Sign In', 'Help', and 'English'. Below this, the 'Web of Science' logo is on the left, and the 'Clarivate Analytics' logo is on the right. A 'Search' bar is located below the logo. To the right of the search bar, there are links for 'Tools', 'Searches and alerts', 'Search History', and 'Marked List'. The main content area shows search results. On the left, a sidebar indicates 'Results: 1,699 (from Web of Science Core Collection)' and 'You searched for: TOPIC: (Alzheimer's disease blood biomarkers) ...More'. Below this is a 'Create an alert' button. The 'Refine Results' section includes a search box and a filter for 'Open Access (881)'. The main results list shows two entries. The first entry is 'Genome-Wide Integrative Analysis Reveals Common Molecular Signatures in Blood and Brain of Alzheimer's Disease' by Rahman, Md Rezanur; Islam, Tania; Shabjam, Md; et al. It is from 'BIOINTERFACE RESEARCH IN APPLIED CHEMISTRY' Volume: 11 Issue: 2 Pages: 8686-8701 Published: APR 15 2021. The second entry is 'Cerebrospinal fluid and serum D-serine concentrations are unaltered across the whole clinical spectrum of Alzheimer's disease' by Nuzzo, Tommaso; Miroballo, Mattia; Casamassa, Alessia; et al. It is from 'BIOCHIMICA ET BIOPHYSICA ACTA-PROTEINS AND PROTEOMICS' Volume: 1868 Issue: 12 Article Number: 140537 Published: DEC 2020. Both entries have 'Times Cited: 0' and 'Usage Count' links. There are buttons for 'Select Page', 'Export...', and 'Add to Marked List' at the top of the results list. There are also links for 'Full Text from Publisher' and 'View Abstract' for each entry.

Web of Science

Clarivate Analytics

Search

Tools Searches and alerts Search History Marked List

Results: 1,699
(from Web of Science Core Collection)

You searched for: TOPIC: (Alzheimer's disease blood biomarkers)
...More

Create an alert

Refine Results

Search within results for...

Filter results by:

☐ Open Access (881)

Refine

Sort by: Date Times Cited Usage Count Relevance More

1 of 170

Select Page Export... Add to Marked List

1. **Genome-Wide Integrative Analysis Reveals Common Molecular Signatures in Blood and Brain of Alzheimer's Disease**
By: Rahman, Md Rezanur; Islam, Tania; Shabjam, Md; et al.
BIOINTERFACE RESEARCH IN APPLIED CHEMISTRY Volume: 11 Issue: 2 Pages: 8686-8701 Published: APR 15 2021
Full Text from Publisher View Abstract

2. **Cerebrospinal fluid and serum D-serine concentrations are unaltered across the whole clinical spectrum of Alzheimer's disease**
By: Nuzzo, Tommaso; Miroballo, Mattia; Casamassa, Alessia; et al.
BIOCHIMICA ET BIOPHYSICA ACTA-PROTEINS AND PROTEOMICS Volume: 1868 Issue: 12 Article Number: 140537 Published: DEC 2020
Full Text from Publisher View Abstract

Analyze Results
Create Citation Report

Times Cited: 0
(from Web of Science Core Collection)

Usage Count

Times Cited: 0
(from Web of Science Core Collection)

Usage Count

Introduction to Endnote software

- Endnote® is a software developed by ISI Research Soft.
(www.endnote.com)
- This software is used for
 - **Creating bibliography database**
 - **Online search tool for reference database**
 - **Retrieving online references and import data files saved from other online services**
 - Helping in writing manuscript and research papers e.g format citations, cite while you write etc.

Online search menu

Remote Search PubMed MEDLINE at PubMed (NLM)

Search In

Search	In	Contains	And
AIDS	Any Field	Contains	And
knowledge	Any Field	Contains	And
Malaysia	Any Field	Contains	

Search Remote: PubMed MEDLINE at PubMed (NLM) Search Close

Boolean search

Type the appropriate key words with “AND”, “OR” or “NOT”
Additional words in the query can be added in Any Field.
Search findings can also be saved in a file.

E.g: Search for reference library
on *AIDS, knowledge, Malaysia*

Searching...

Confirm Remote Search

Found 12 references.

Retrieve References From: 1 through 12

☐ Discard the previously retrieved references?

OK Cancel

This search retrieves 12 references. If too many, you can decide how many to retrieve at one time. E.g from “1 – 6” then from “7 – 12”

To sum up

- *Knowing how to search is more than half the battle. The other half is knowing what to search, that is, which databases.*
- *The best rule is practice makes perfect . . . do it yourself, do it often, and don't depend on research assistants alone!*

Injury Prevention 2003;9:103–104

Sources

- Steinbrook, R. Searching for the Right Search — Reaching the Medical Literature. Downloaded from www.nejm.org at CAUL on Feb.7, 2009.
- Soltoggio, A. Organised and Systematic Literature Search. Downloaded from <http://www.cs.bham.ac.uk/~aqs>
- Gore, G. Searching the medical literature. Downloaded from injuryprevention.bmj.com on 7 February 2009
- Matthew et al, 2008. *FASEB J.* 22, 338–342 (2008)
- <http://www.rgu.ac.uk/library/howto/page.cfm?page=25990>
- <http://www.nursingtimes.net/nursing-practice-clinical-research/how-to-conduct-an-effective-and-valid-literature-search/217252.article>



EVALUATING THE LITERATURE

- Titles (article/books)
- Abstract (article)
 - Study purpose, research strategy, findings, conclusion
- Introduction (articles/book)
- Table of contents (books)
- First chapter (books)